Mechanical-Hydraulic Modeling of an Embankment Mower in Simscape



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Outline

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- Mechanical Subsystem
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Introduction and Motivation

- The Mulag FME 500 is a heavy hydraulic manipulator that works in stochastic environments.

The Problem

- Accurate simulation is difficult because of bidirectional coupling between the domains e.g. hydraulic forces drive the mechanics and mechanical loads alter fluid properties.
- Standard kinematics fail to capture „stiff“ oscillatory behaviors.

The Goal

- Develop a high fidelity physically consistent Digital Twin in Simscape that captures these nonlinearities to support future safety verification and control development.

Theoratical Background

Hydraulic Nonlinearities

The system is governed by nonlinearities such as ***bulk modulus*** and stribeck friction.

Modeling Philosophy

The approach is physical modeling. Simscape is used to solve the rigid multibody mechanics and hydraulic networks simultaneously ensuring energy conservation across dormains.

Simscape versus Robot Engines

Feature	Simscape	Robot Engines
1. Coupling Strategy	Unified and bidirectional	Decoupled and assumes ideal actuation
2. Actuator Fidelity	High (physical)	Low (kinematic)
3. Solver Type	Implicit (stiff)	Explicit (fast)
4. Component Detail	Physical networks	Abstract constraints

Mechanical Subsystem (1)

Workflow



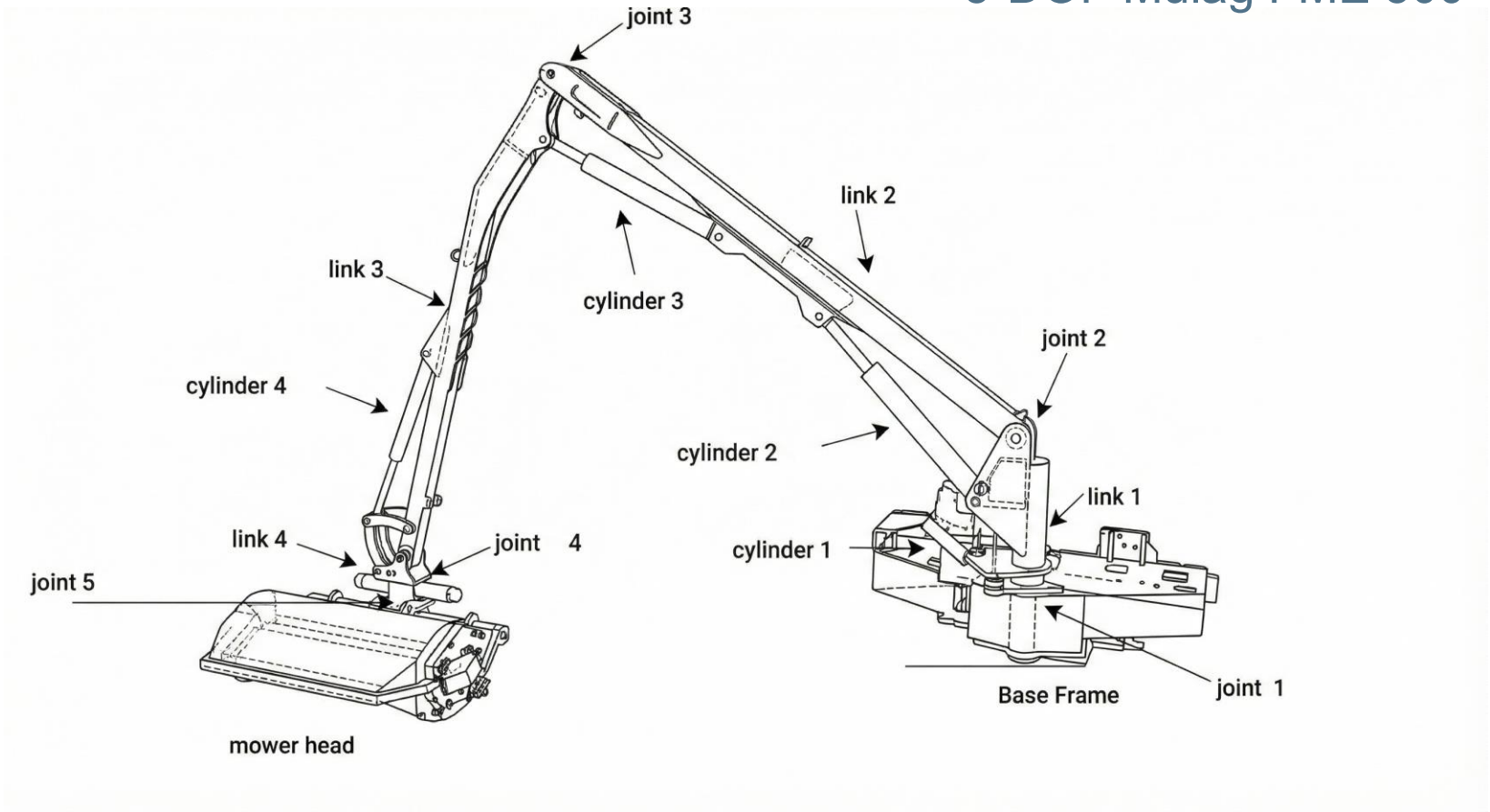
- Automated Pipeline CAD to URDF to ensure simulation captures mass distribution and cross couplings inertia terms (I_{xy}, I_{yz}) are preserved.

- Simscape Integration and Closed-loop kinematic of cylinders $(S_{base} - P - C_{rod})$ and frame alignment. Assign masses and inertias of rigid bodies from geometry.

- Force –to-torque coupling i.e. converting cylinder (F_h) to revolute joint torque using the kinematic Jacobian $J_{cyl}(q)$: $\tau_{act} = J_{cyl}^T(q)F_h$

Mechanical Subsystem (2)

- 5-DOF Mulag FME 500

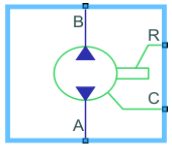


Hydraulic Subsystem (1)

Circuit Design

- Fixed-displacement pump, gas-charged accumulator, 4/3 proportional directional control valves that drive the double-acting cylinders.
- Dynamic response of the hydraulic cylinder is governed by the bulk modulus of working fluid.

Pump

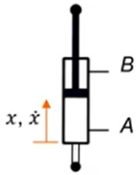


Fixed-Displacement Pump (IL)1

- Constant flow generation.

$$Q_p = D_\rho \omega$$

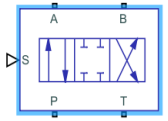
Cylinders



- $A \neq B$
- Dead volumes
- Stribeck friction and internal leakage

Valve

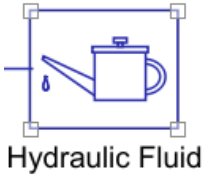
- Flow is nonlinear and follows orifice equation



4-Way 3-Position Directional Valve (IL)

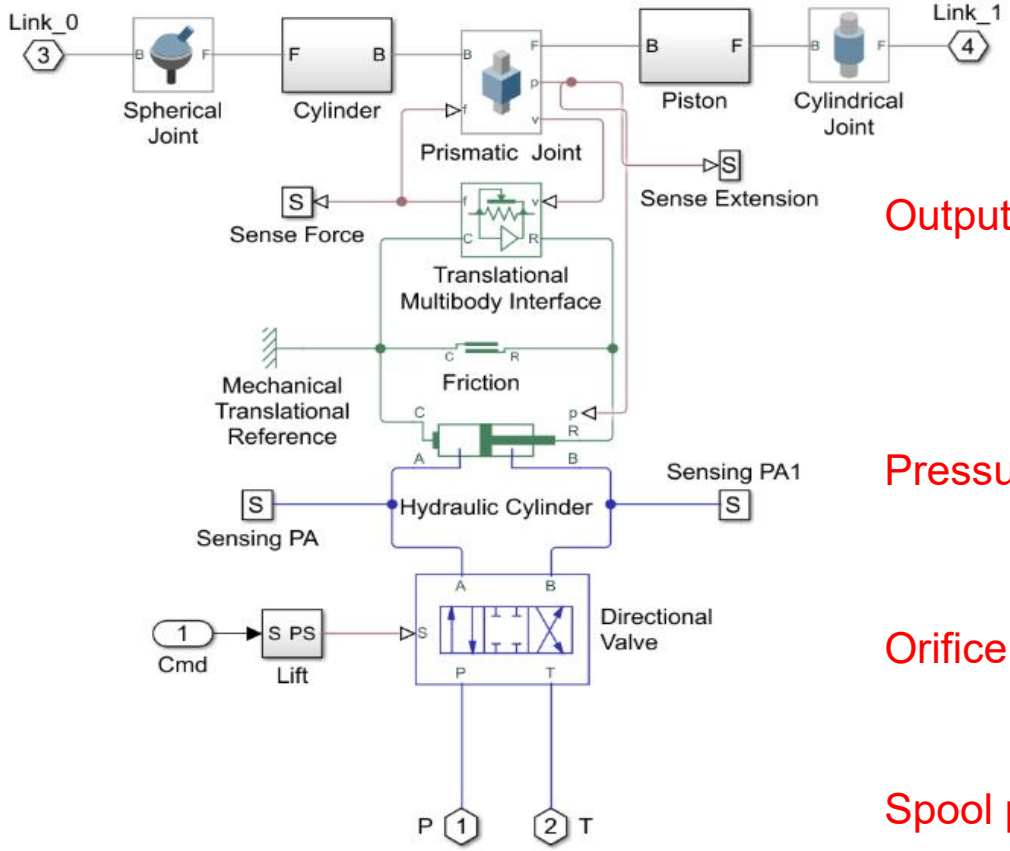
$$\dot{x}_v = \frac{1}{\tau_v}(u_{cmd} - x_v)$$

Fluid Properties



- E_{eff}
- Entrained Air
- Bidirectional coupling

Hydraulic Subsystem (2)



- Electro-hydraulic actuator subsystem

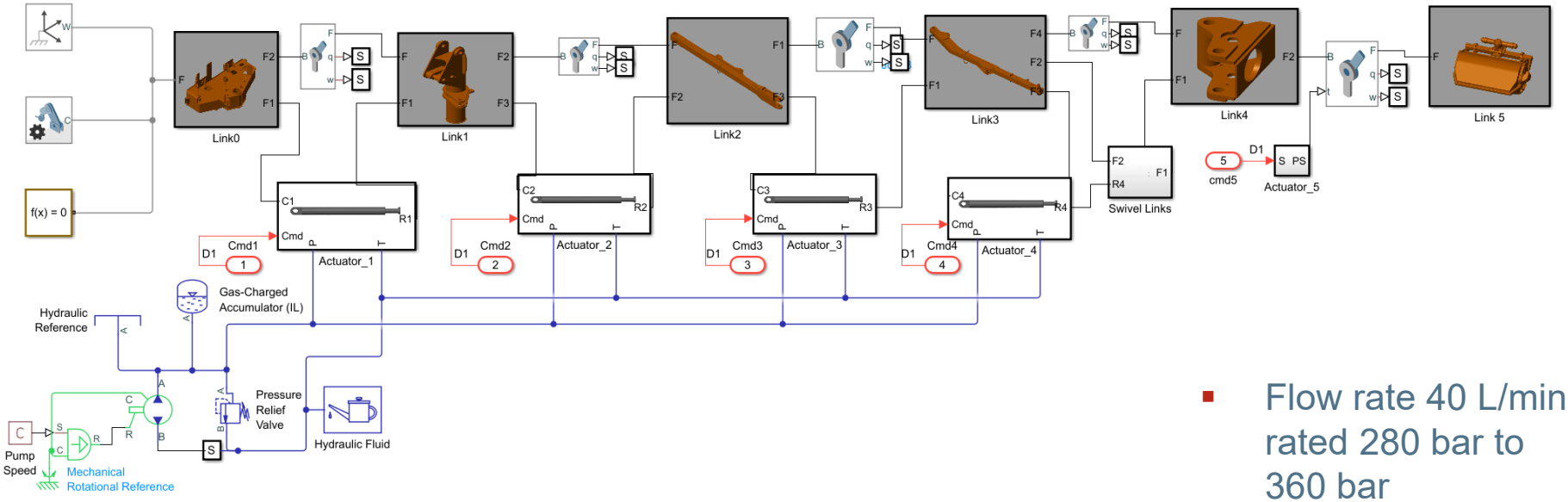
Output Force: $F_h = p_1 A - p_2 B - F_{fric}$ (4)

Pressure Buildup $\dot{p}_1 = \frac{E_{eff}}{V_1(x)}(Q_1 - A\dot{x} - Leakage)$ (3)

Orifice Equation: $Q = C_d \cdot w \cdot x_V \sqrt{\frac{2}{\rho}\Delta p}$ (2)

Spool position: $x_v = u_{cmd} - \tau_v \dot{x}_v$ (1)

Integrated System



- Flow rate 40 L/min rated 280 bar to 360 bar

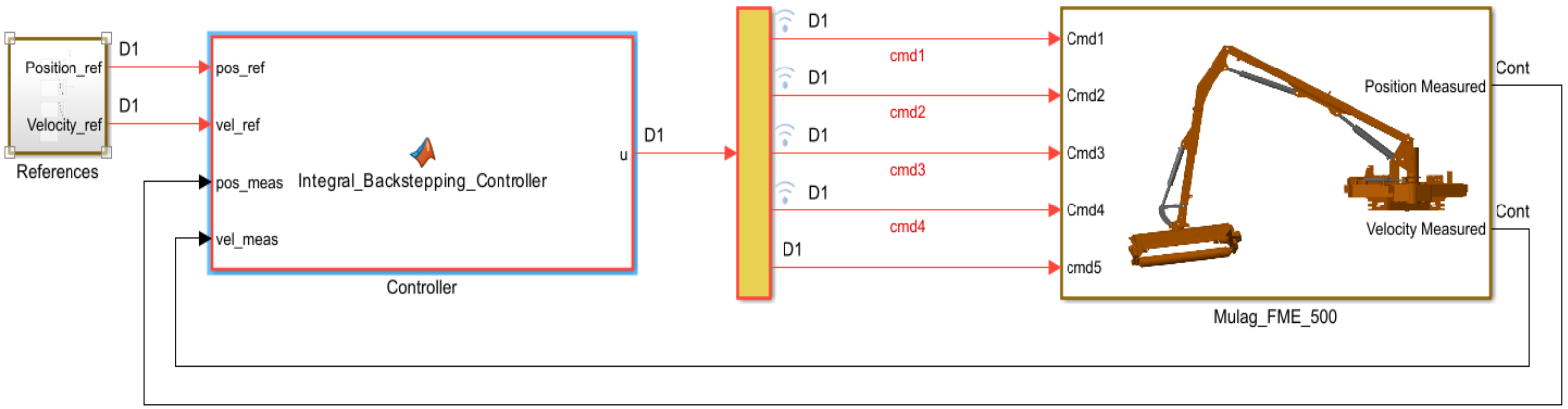
- The system is stiff due to the mix of fast pressure transients and slow boom motion.
- Variable-step implicit solver is used to resolve compliance without the global step sizeo [ode23t]

Control Design

- The controller serves as a validation instrument to validate the plant.
- A decentralized **Integral Backstepping Controller** is used to stabilize error dynamics.

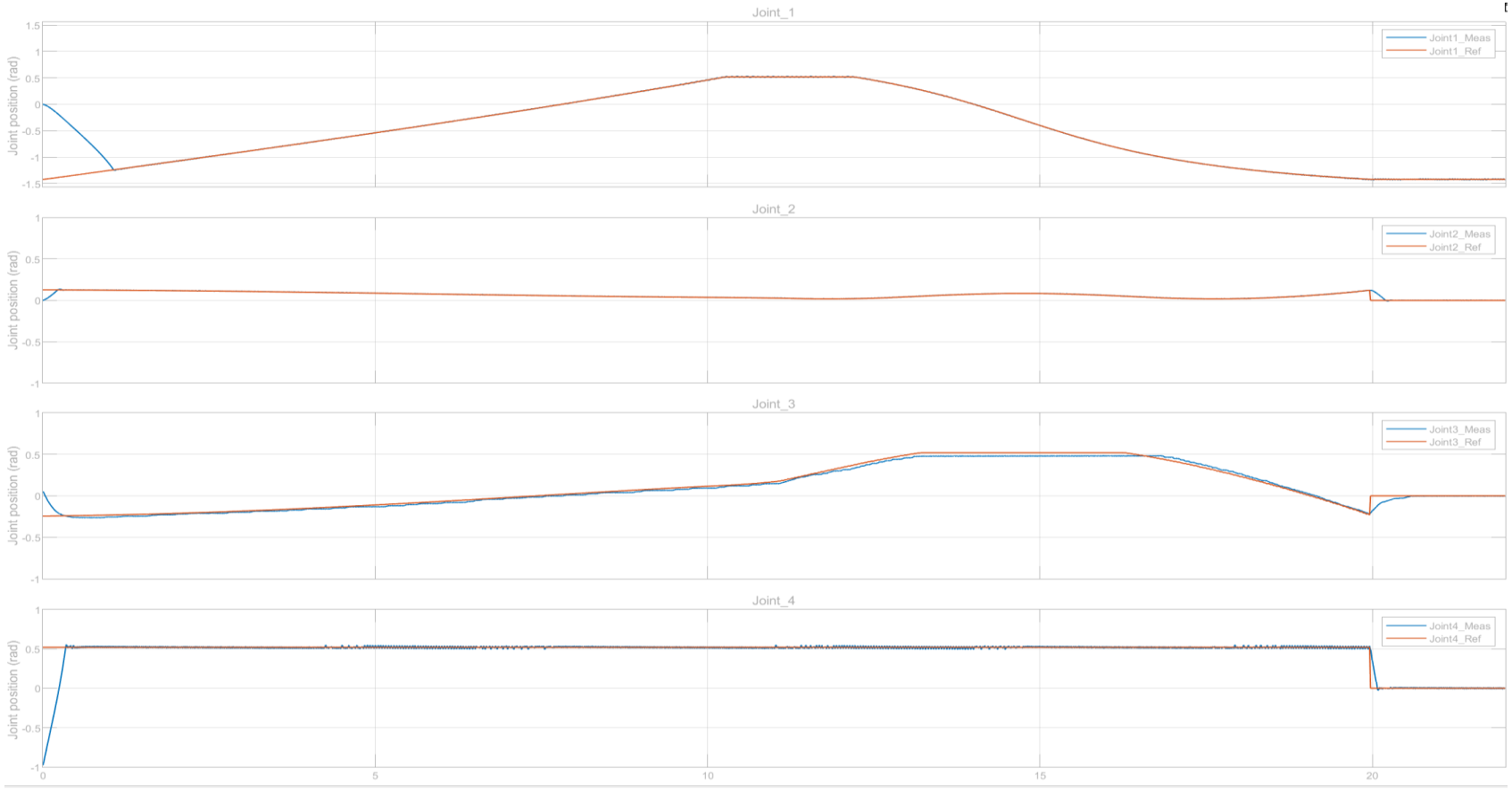
1. Position error: $z_1 = q_{ref} - q_{meas}$
2. Virtual velocity: $\alpha = \dot{q}_{ref} + c_1 z_1$
3. Velocity error: $z_2 = \alpha - \dot{q}_{meas}$
4. Integral Action: $Z_{int} = \int z_2 \, dt \approx \sum z_2 \cdot T_S$
5. Control Law: $v = z_1 + c_2 z_2 + K_{int} Z_{int}$

$$\boldsymbol{u = \frac{v + U_{fric} \cdot sgn(z_2)}{b_{sys}}} \quad , b_{sys} \in [-1,1]$$

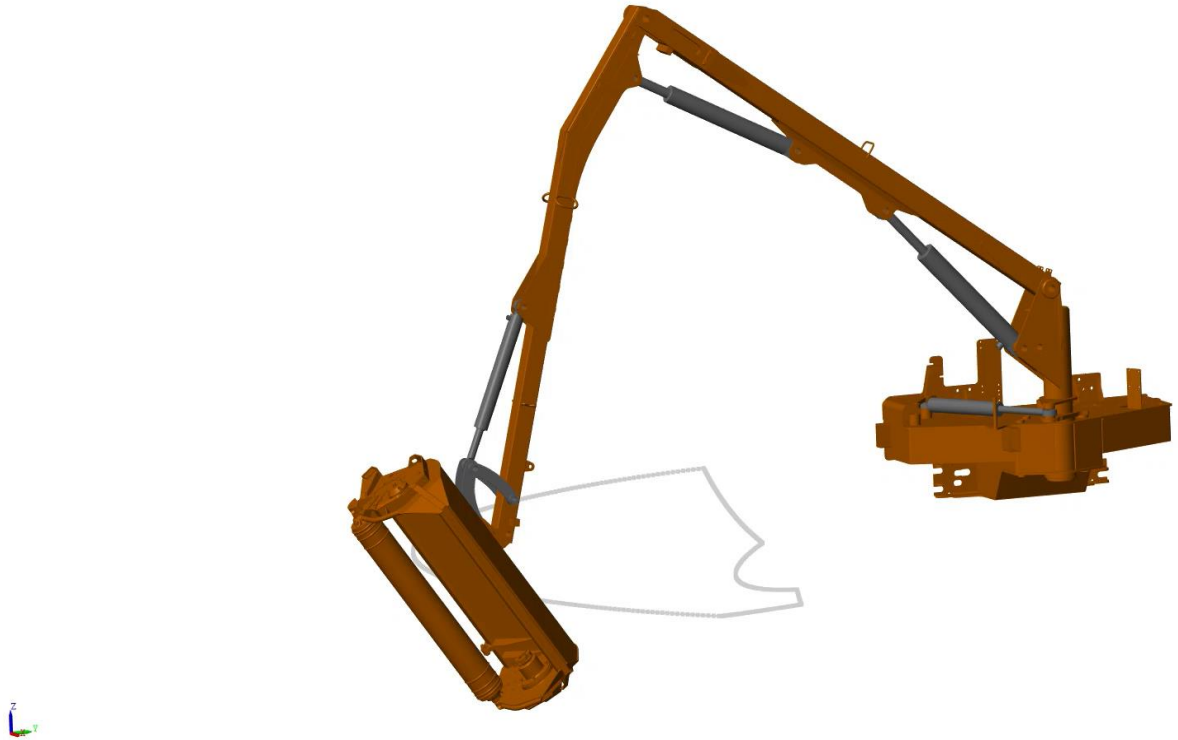


Control Design

Joint Space Control



Simulation



- With the control architecture in place, the mower head was commanded to follow a closed-loop reference path in the task space.

Validation Strategy (Hydraulic Actuation Delay)

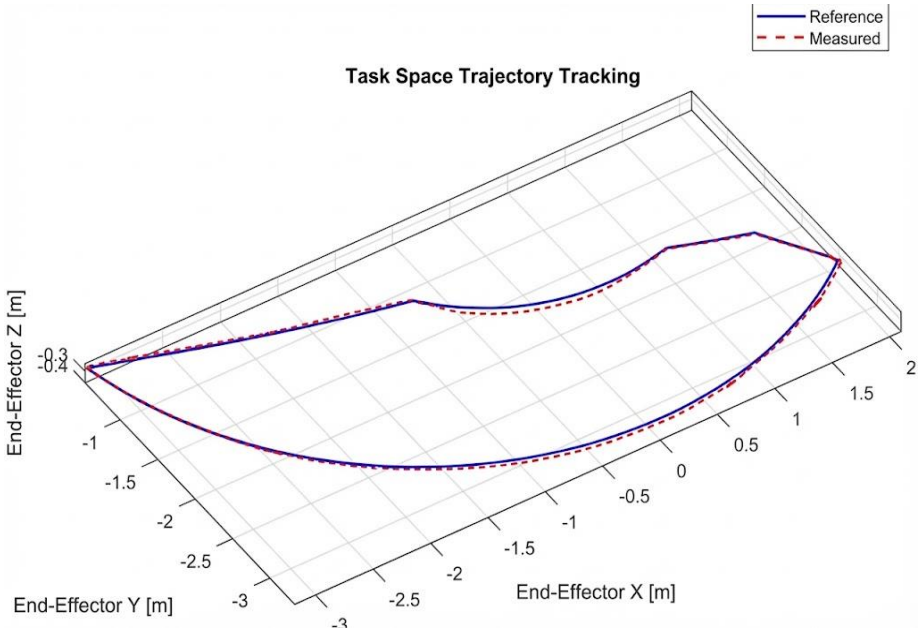
- Signal propagation is tracked through three stages:

Valve command(T_1) ⟶ **Pressure Buildup**(T_2) ⟶ **Joint Motion**(T_3)

- To avoid noise, start time is defined for each event the moment it reaches 50% of it's maximum value.
- The primary metric is $(\boldsymbol{\tau_v \rightarrow joint})$ valve to joint delay (total time elapsed from valve and when arm starts to move i.e. $(\boldsymbol{T_3 - T_1})$).
- The system score the **delay index** $(\boldsymbol{\tau_{RMS}})$ aggregates the delays of all the 4 joints to score overall sluggishness of the machine.

$$\tau_{RMS} = \sqrt{\frac{1}{4}\sum_{j=1}^{4}\left(\tau_{v\rightarrow joint}^{(j)}\right)^2}$$

Validation Strategy (Task Space Tracking Error)



- The instantaneous Euclidean distance between reference path $\boldsymbol{p_{ref}(t)}$ and mower head position $\boldsymbol{p_{meas}(t)}$ is quantified.

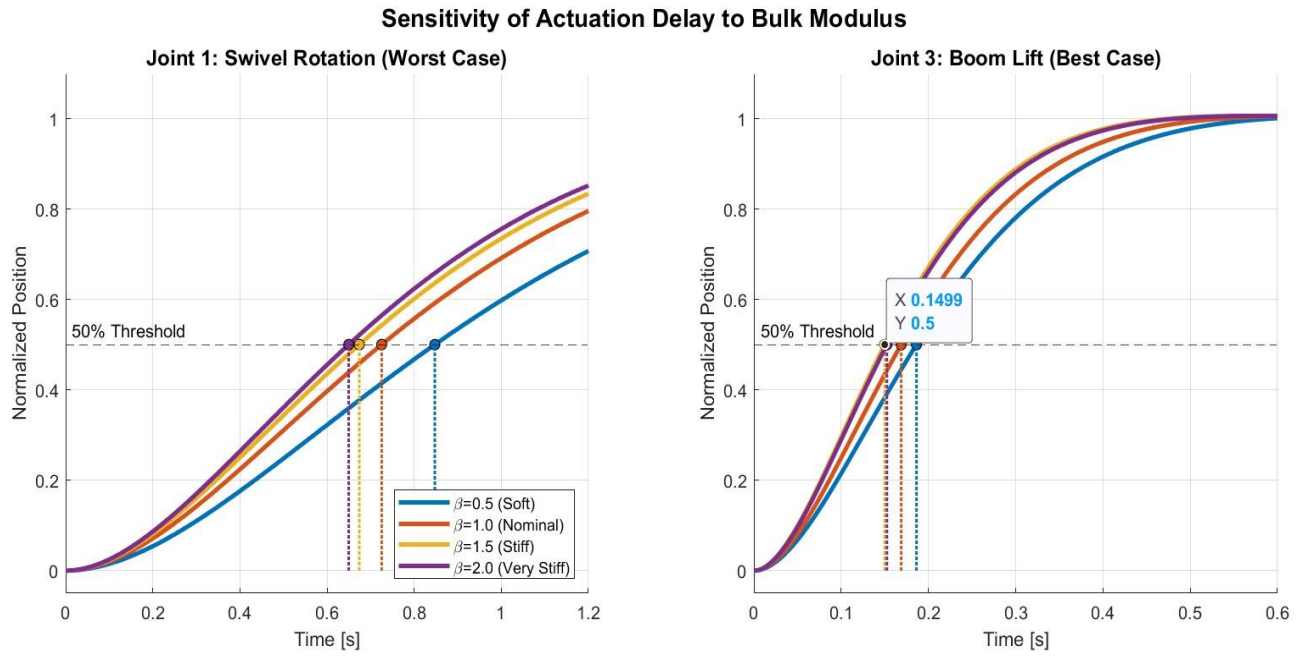
- The global score is summarized as the **Root Mean Square (RMS)** of **Error** denoted as $\boldsymbol{E_{RMS}}$ given by formula:

$$E_{RMS} = \sqrt{\frac{1}{T}\int_{t_{\text{start}}}^{t_{\text{end}}} \mathrm{e}^2(t)\,\mathrm{d}t}$$

Results and Discussion

$\beta[GPa]$	$\tau_v(1) \rightarrow joint$ [ms]	$\tau_v(3) \rightarrow joint$ [ms]	$\tau_v(4) \rightarrow joint$ [ms]	τ_{RMS} [ms]
0.5	847.4	186.3	202.1	445.4
1.0	725.3	168.8	206.4	386.4
1.5	674.0	149.9	215.8	361.7
2.0	649.7	152.4	221.6	351.6

- Valve-to-joint 50% delays and delay index for different bulk moduli

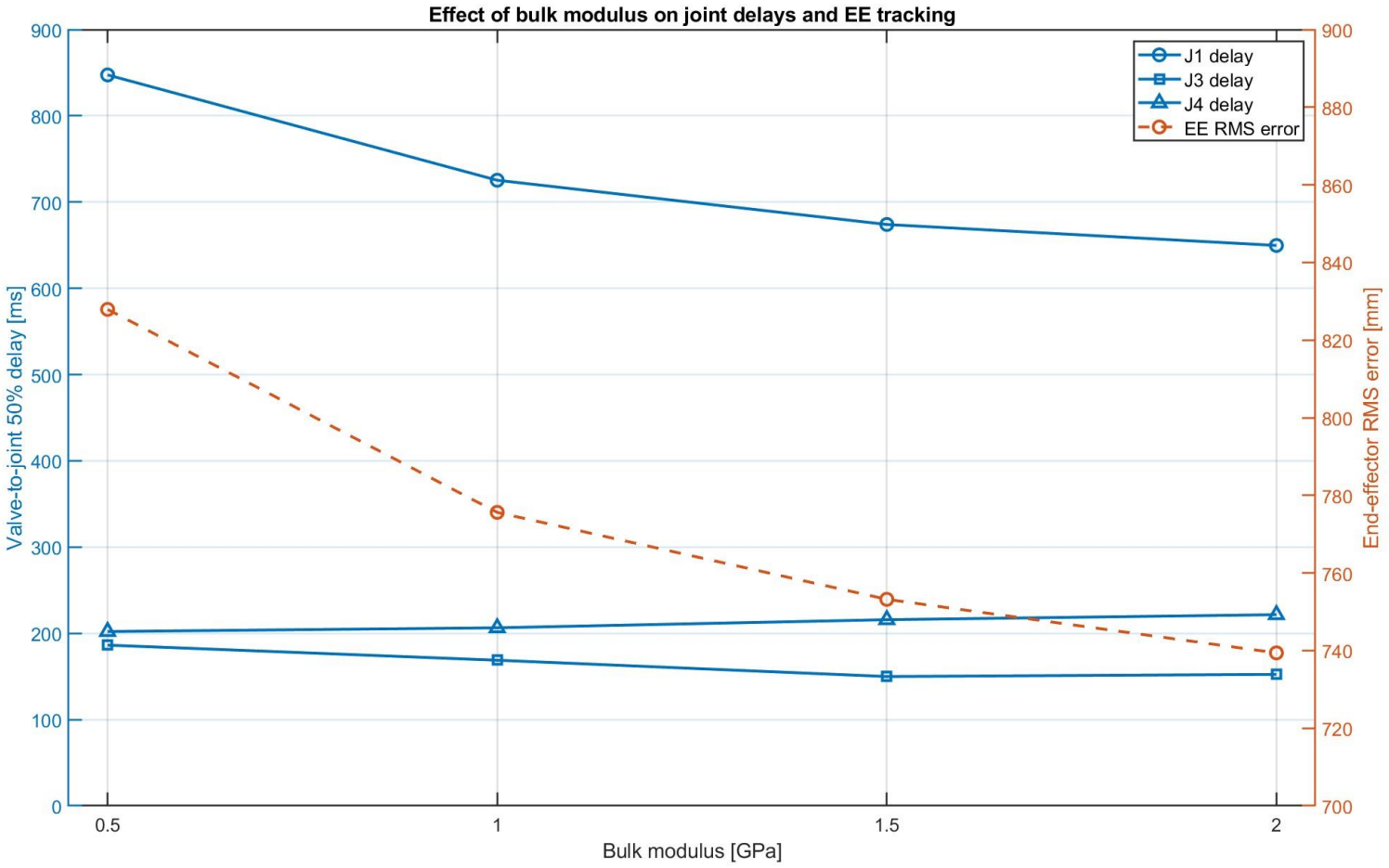


Results and Discussion

- End effector Accuracy to Bulk Modulus

$\beta[GPa]$	E_{RMS} [mm]	E_{mean} [mm]
0.5	827.9	211.3
1.0	775.6	215.4
1.5	753.2	229.1
2.0	739.4	228.8

Results and Discussion



Conclusion & Future Work

- A model that bridges the gap between rigid multibody mechanics and nonlinear hydraulic dynamics was successfully developed.
- The simulaton captures the bidirectional coupling i.e. actuator forces drive the structure while structural loads induce transients.
- The results from valiadtion confirm the model physical intergrity specifically the ability to replicate „stiff“ yet oscillatory nature of hydraulic machinery and sensitivity of delays to fluid compressibility.
- This validated model will serve as a digital twin for developing data-driven control methods.
- The goals are to learn the complex nonlinear hydraulic-mechanical coupling then develop advanced control architectures capable hydraulic delays compensation.

Questions